

Trajectory Analysis of Intestinal T-Cell Compartments in Crohn's Disease

Slingshot-based pseudotime inference in IEL and LPL compartments, with
condition-descriptive diagnostics

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Executive Summary

This report describes the trajectory inference analysis performed on the intraepithelial lymphocyte (IEL) and lamina propria lymphocyte (LPL) compartments of the Jaeger et al. 2021 dataset, using Slingshot as the sole trajectory inference engine. Unlike the pseudobulk DESeq2, CellChat, and LIANA analyses conducted elsewhere in this project, the trajectory analysis does **not** attempt a formal statistical comparison of Crohn's disease (CD) versus Control along pseudotime. This choice was made after diagnostic inspection of both compartments revealed condition-associated cell-type composition imbalance so severe that most cell states are effectively condition-exclusive, making formal cross-condition trajectory comparisons poorly informative for many lineages and highly sensitive to donor composition, given the strong condition-associated cell-state imbalance and the $n = 4$ donor design.

Instead, Slingshot was fit once per compartment on the pooled (condition-agnostic) dataset, producing a single descriptive map of cell states. Disease condition is then overlaid onto this map purely descriptively — via condiments' `imbalance_score`, pseudotime density by condition, and compositional profiles along pseudotime bins — without any formal hypothesis test. This is a deliberate and explicitly justified methodological departure from the condiments/tradeSeq framework introduced in earlier drafts of this analysis, and is documented as such throughout.

Two substantive findings emerged during quality control that materially affect interpretation, and one of them directly resolves an open question from the bulk RNA-seq cross-validation report:

1. **IEL** — a shared trajectory trunk connects CD4-labeled and CD8-labeled cell states without a clean, monotonic co-receptor reprogramming gradient on canonical markers (CD4, CD8A, CD8B, RUNX3, ZBTB7B); this trunk should be read as a region of transcriptional similarity rather than a literal CD4→CD8 developmental conversion.
2. **LPL** — several CD-associated populations, including the FOXP3+ activated Treg population whose increased relative abundance in CD had been flagged as an unresolved discrepancy with Jaeger et al. (2021) in the bulk RNA-seq report, are contributed almost exclusively by a single CD donor, not fully resolved by Harmony integration. This finding provides a donor-level explanation for the observed discrepancy in this dataset: the apparent increase in LPL Treg abundance is driven almost entirely by a single CD donor and therefore cannot be interpreted as a generalizable disease-associated shift.

Analytical Workflow

The analytical pipeline was applied independently to each compartment (IEL and LPL), following an identical structure:

- SingleCellExperiment (SCE) construction from Seurat v5 objects, with raw counts pulled from the joined, whole-object layer to avoid multi-layer ambiguity, and explicit assignment of `counts/logcounts` assays.
- Condition-imbalance diagnostic (`condiments::imbalance_score`) computed on the full compartment UMAP, prior to any trajectory fitting.
- Removal of non-lymphoid / technically negligible populations (contaminant epithelial and myeloid clusters) from the cells used for trajectory fitting.
- Slingshot lineage inference (`getLineages`) on the PCA embedding, using cell-type labels as cluster input, an explicitly verified biological root cluster, and `omega = TRUE` to avoid forcing geometrically distant, biologically unrelated clusters onto a single connected tree.
- Principal curve fitting (`getCurves`), pseudotime and curve-weight extraction (`slingPseudotime`, `slingCurveWeights`).
- Descriptive, condition-agnostic characterization of pseudotime: per-lineage visualization, condition density overlays, and compositional profiles along pseudotime bins, plus descriptive marker-trend plots (binned means, no model fit).
- No `condiments` statistical tests (`topologyTest`, `progressionTest`, `differentiationTest`, `fateSelectionTest`) and no `tradeSeq` model fitting (`fitGAM`, `associationTest`, `startVsEndTest`, `conditionTest`) were run as part of the final reported workflow — see the rationale below.

Methodological Rationale: Why No Formal CD vs Control Trajectory Testing

The `condiments` framework (Roux de Bézieux et al., *Nat Commun* 2024) is designed to formalize the comparison of a shared trajectory across two conditions: does topology differ, does pseudotime progression differ along a shared lineage, does fate selection differ at a branch point. This framework presupposes that most or all trajectory states are meaningfully populated by both conditions. Diagnostic inspection of both IEL and LPL (see per-compartment sections below) showed that this precondition is not met: the large majority of annotated cell states in both compartments are condition-exclusive or near-exclusive (90–99% of cells from a single condition), a compositional pattern already established and characterized elsewhere in this project via the CellChat/LIANA validity-tier framework (fully comparable / imbalanced / CD-emergent / Control-dominant).

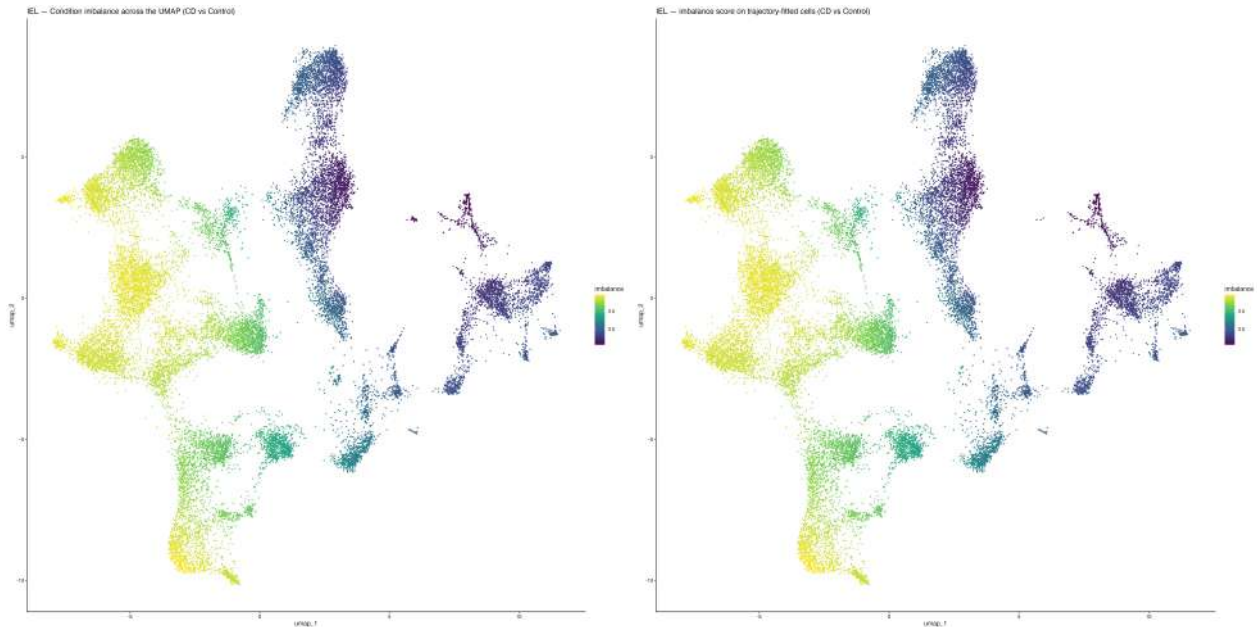
Applying `condiments`' or `tradeSeq`'s condition-aware tests to trajectory branches that are condition-exclusive would not test whether the same biological process proceeds at different kinetics between conditions; it would merely re-derive, with an inferential p-value, the compositional imbalance already documented via simple contingency tables. Combined with the project's $n = 4$ donor design (2 CD / 2 Control per compartment, never modeled with a donor-blocking term in any analysis layer of this project), the effective replication available for a genuine cross-condition kinetic comparison is negligible on most branches. For this reason, condition is treated throughout this analysis as a **descriptive covariate** overlaid on a single, condition-agnostic trajectory map, rather than as an experimental factor in a formal statistical model — the same logic already applied to CD-emergent and Control-dominant populations in the CellChat/LIANA communication analysis of this project.

A single formal check — `condiments::topologyTest_multipleSamples`, which accounts for donor-of-origin via a `Samples` argument rather than treating every cell as an independent replicate — was considered as an optional, one-time documented consistency check rather than a decision gate, but was ultimately not required to justify the descriptive approach given how unambiguous the compositional evidence already was in both compartments.

Results — IEL Compartment

Composition Imbalance

Cell-type composition in IEL is severely imbalanced by condition (global ratio: 44% CD / 56% Control). The large majority of annotated populations are condition-exclusive or near-exclusive: CD-dominant examples include FOXP3+ IL2RA+ Treg (79 CD / 0 Control), Terminal effector CD8 T cells (529 / 0), TH17-like CD4 T cells (1619 / 2), and Cytotoxic T cells (2113 / 11); Control-dominant examples include IL7R+ memory-like CD8 T cells (0 / 1481), T cells / NK-like cytotoxic T cells (1 / 2287), and “Naïve CD4 T cells” (4 / 1935 — not to be confused with the marker-verified “Naïve CD4” root cluster below, a separate, correctly-spelled label in the annotation).



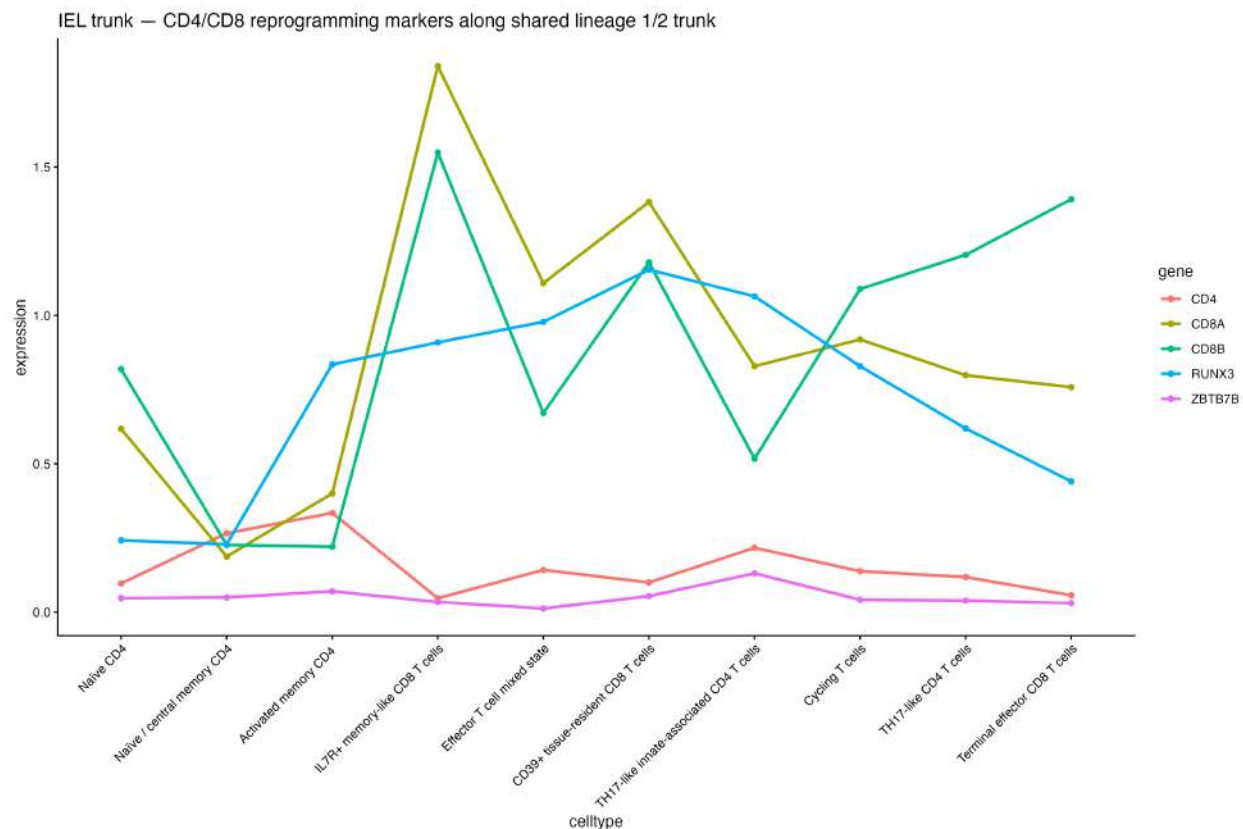
Root Cluster Selection

Two candidate labels (“Naïve CD4” and “Naïve / central memory CD4”) were both confirmed as genuinely quiescent/naïve by mean expression of CCR7, TCF7, SELL, and LEF1 across all IEL clusters, prior to selecting a root. “Naïve CD4” scored highest on all four markers (CCR7 1.40, TCF7 1.19, SELL 1.56, LEF1 0.66) and was selected as the Slingshot root (`start.clus`). A third, superficially similar label (“Naïve CD4 T cells”) was explicitly ruled out despite its name, showing near-zero CCR7/SELL (0.009 / 0.001) — a reminder that cluster labels were not taken at face value for this decision. No IEL population showed a comparable naïve-like profile for the CD8 lineage, a limitation carried into the topology discussion below.

Trajectory Topology

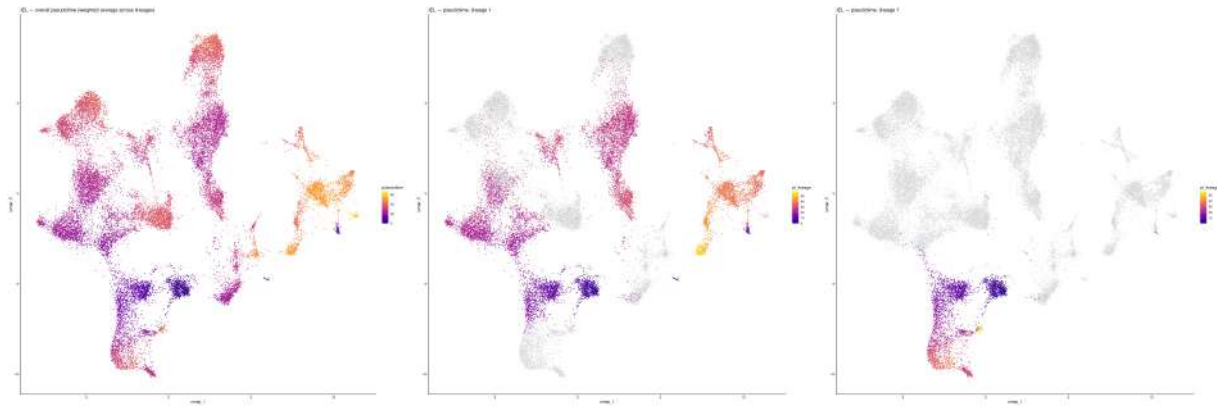
Slingshot (`getLineages`, PCA embedding, `omega = TRUE`, `start.clus = "Naïve CD4"`) recovered 7 lineages, all sharing “Naïve CD4” as their root — consistent with, though not independently re-confirmed via an explicit `start.given` metadata check the way it was for LPL (see below). `omega = TRUE` was explicitly confirmed active (`omega_scale = 1.5`) via `metadata(lineages_iel)$slingParams`. Lineages 1–2 share a long common trunk that alternates between CD4-labeled and CD8-labeled states (Naïve CD4 → Naïve/CM CD4 → Activated memory CD4 → IL7R+ memory-like CD8 → Effector mixed → CD39+ tissue-resident CD8 → TH17-like innate CD4 → Cycling → TH17-like CD4 → terminal CD8/Treg). This trunk persisted

unchanged after `omega = TRUE` was applied, indicating the geometric distance between these CD4- and CD8-labeled clusters in PCA space does not exceed the omega threshold ($1.5 \times$ median MST edge length) — i.e. this adjacency is a property of the data, not an artifact the omega correction could have removed.



This adjacency was investigated against the documented phenomenon of local CD4→CD8 lineage recommitment in gut intraepithelial lymphocytes (Mucida et al., *Nat Immunol* 2013, 14:281–289; Reis et al., *Nat Immunol* 2013, 14:271–280; reviewed in Park, Moon & Lee, *BMB Reports* 2016, 49:11–17), which would predict a monotonic decline in CD4/ZBTB7B and a monotonic rise in CD8A/CD8B/RUNX3 along the trunk. The observed pattern does **not** support this: CD4 is uniformly low throughout the trunk (a known technical limitation of CD4 mRNA detection in scRNA-seq, not by itself informative), ZBTB7B remains near-zero throughout without ever being elevated at the presumptive root, and CD8A/RUNX3 rise through the middle of the trunk but decline toward the terminal state rather than continuing to increase. The trunk is therefore reported as a region of shared transcriptional similarity — consistent with the general convergence of gut IEL states on partial CD8 /RUNX3 programs regardless of nominal CD4/CD8 identity — rather than as evidence of a literal, ordered lineage-conversion process.

Pseudotime

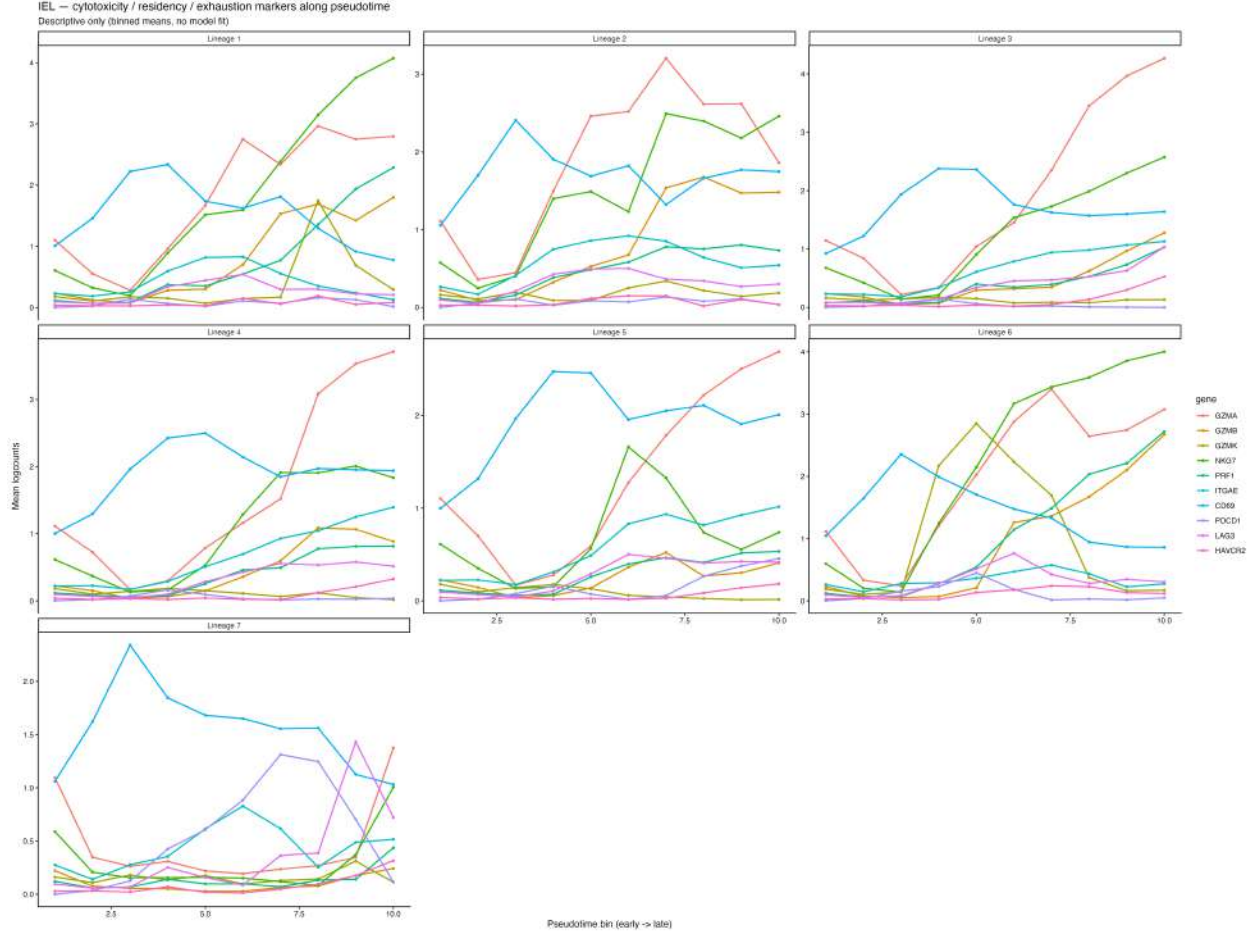


Pseudotime gradients are spatially coherent across all 7 lineages, with no evidence of embedding breakdown. Lineages 5–7 are short (3–4 nodes) and terminate in small, spatially isolated populations; their pseudotime ranges, while wide, are compressed into narrow UMAP regions and should be interpreted with more caution than the longer lineages.

Descriptive Condition Overlay

Condition is overlaid on the IEL trajectory purely descriptively, via pseudotime density by condition and a compositional profile (proportion CD) across ten pseudotime bins per lineage (figures produced by the pipeline as `Pseudotime_distribution_condition.png` and `CompositionProfile.png` for the IEL compartment; see project figure archive — these particular files were superseded on-disk by same-named LPL exports and are referenced here by content rather than re-embedded).

Lineages 1, 2, and 3 show a consistent, non-monotonic (“U-shaped”) compositional pattern: CD dominates the earliest pseudotime bin (an inherited effect of the CD-skewed “Naïve CD4” root, 678 CD / 8 Control), Control dominates the intermediate bins, and CD dominates again toward the terminal bins. Lineage 7 (terminating in CXCL13+ TFH-like T cells, 94.9% Control) is the most extreme case: CD is confined almost entirely to the first pseudotime bin, consistent with its near-total Control dominance in the underlying cell-type table.



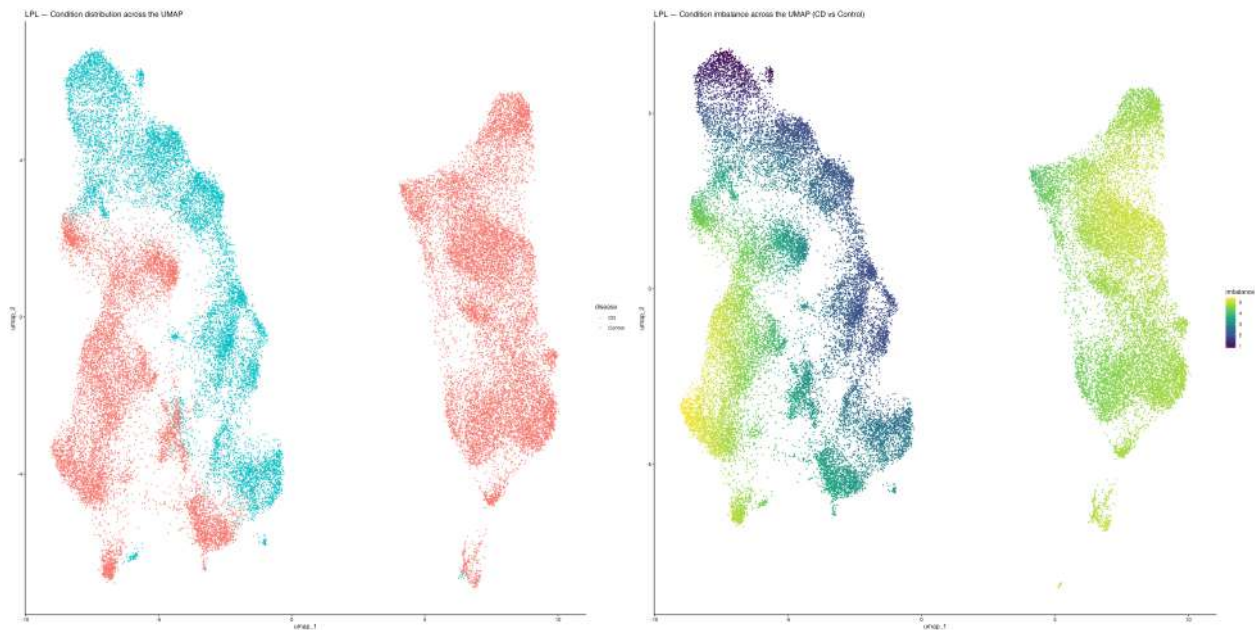
IEL — Compartment-Specific Limitations

- No naive-like CD8 population was identified in IEL by canonical markers, meaning the Slingshot root is necessarily CD4-lineage; any CD8 branch is therefore rooted through the shared trunk rather than through an independently verified CD8 origin.
- The shared CD4/CD8 trunk (Lineages 1–2) should not be read as evidence of a literal reprogramming trajectory; see marker verification above.
- Lineages 5–7 are short and terminate in small, spatially compressed populations; pseudotime within them is more susceptible to noise than in the longer lineages.
- As throughout this project, all compositional and descriptive condition comparisons are limited by the $n = 4$ donor design (2 CD / 2 Control), with no donor-blocking term used in any analytical layer.

Results — LPL Compartment

Composition Imbalance

As in IEL, the large majority of LPL populations are condition-exclusive or near-exclusive. Notable examples: FOXP3+ activated Treg (2205 CD / 0 Control), NK-like cytotoxic T cells (296 / 0), TRM-like CD8 cytotoxic T cells (204 / 0), Activated CD4 memory T cells (5578 / 1), Activated CD4 T cells stress response (15 / 2628), and T follicular helper (24 / 2000).

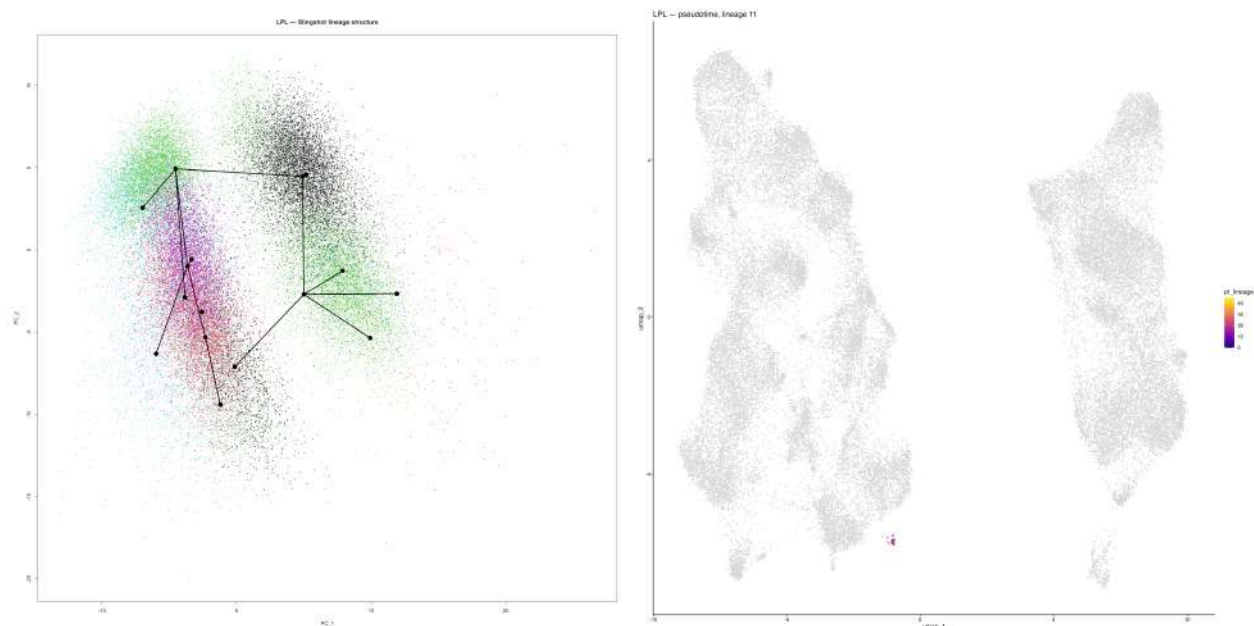


Root Cluster Selection

“Naive CD4 T cells” was used as the Slingshot root (`start.clus`), confirmed via `start.given = TRUE`. Unlike IEL, this selection was **not** independently cross-checked against canonical naive markers (CCR7/TCF7/SELL/LEF1) prior to fitting; it is used here because it is the only unambiguously naive-labeled population in the LPL annotation set and its label did not require the same disambiguation that IEL’s near-duplicate naive labels did. This is noted as a lighter-touch verification than was applied in IEL.

Trajectory Topology

Slingshot (`omega = TRUE`) recovered 11 lineages. Ten of these (Lineages 1–10) share “Naive CD4 T cells” as a common root and remain connected in a single tree; `omega` correctly isolated only Lineage 11 (root and end both “Exhaustion-like CD4 memory T cells”, $n = 36$ cells total, 3 CD / 33 Control) as an independent sub-tree, which should be treated as non-conclusive given its small size and spatial isolation.



Donor Effect on Lineages 1–10 (Critical Finding)

Visual inspection of the LPL UMAP revealed two spatially disconnected point clouds (“islands”), with the smaller, right-hand island almost entirely composed of CD cells. Because `omega = TRUE` did not separate this island into an independent sub-tree, Lineages 1–10 extend across both islands via PCA-space adjacency that is not visually apparent in the UMAP projection.

Donor-level inspection resolved this decisively: of the four donors, donor **1818 (CD)** contributes 15,247 of its 15,264 total cells (99.9%) to the right-hand UMAP island, while the other three donors are almost entirely confined to the left-hand island. Cross-referencing with cell-type composition shows this single donor accounts for the overwhelming majority of several of the compartment’s most CD-skewed populations:

Cell type	Cells from donor 1818	Total cells (compartment)
FOXP3+ activated Treg	2204	2205
Activated CD4 memory T cells	5578	5579
NK-like cytotoxic T cells	296	296
TRM-like CD8 cytotoxic T cells	204	204

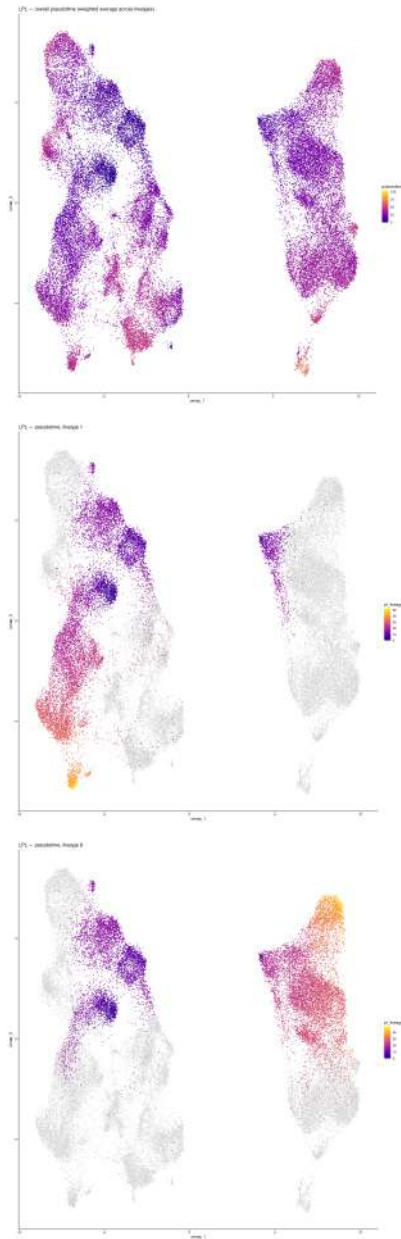
These populations are therefore contributed almost exclusively by a single donor, not resolved by Harmony integration, rather than being shared, reproducible CD-associated states. With $n = 2$ CD donors in this compartment, a population contributed at this level by one donor is statistically indistinguishable from a donor-specific effect.

Connection to the Bulk RNA-seq Report: The Treg Discrepancy, Resolved

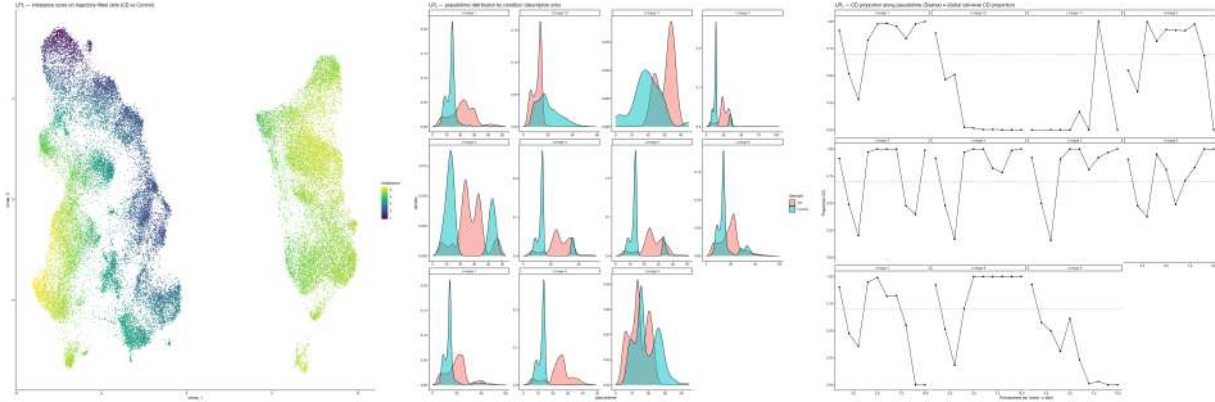
The bulk RNA-seq cross-validation report (GSE193677 analysis) flagged an unresolved discrepancy: this project’s pseudobulk/scRNA-seq compositional analysis found an increased relative Treg proportion in CD (LPL: 0.12 CD vs 0.05 Control), directly contradicting Jaeger et al. (2021), who reported reduced Treg proportions in inflamed CD tissue. That report explicitly declined to resolve the discrepancy, attributing it tentatively to possible differences in Treg subtype definition or sampling between studies.

The donor-level finding above provides a direct, mechanistic resolution: **2204 of the 2205 FOXP3+ activated Treg cells in this project's LPL compartment originate from a single CD donor (1818)**. The apparent CD-associated increase in Treg proportion is therefore not distinguishable from a donor-specific effect and should not be interpreted as a generalizable, disease-associated compositional shift. This finding should be incorporated into the manuscript's discussion of the Treg discrepancy in place of the previously open-ended language.

Pseudotime



Descriptive Condition Overlay



LPL — Compartment-Specific Limitations

- A subset of Lineages 1–10 (those reaching the donor-1818-dominated UMAP region, including at minimum the lineages terminating in FOXP3+ activated Treg, NK-like cytotoxic T cells, TRM-like CD8 cytotoxic T cells, and passing through Activated CD4 memory T cells) cannot be distinguished from a single-donor effect and must not be reported as generalizable CD-associated trajectory findings.
- Root selection (“Naive CD4 T cells”) was not independently marker-verified as it was in IEL; this is a lower-rigor step than the equivalent IEL decision.
- Lineage 11 is based on 36 cells total and is non-conclusive.
- As in IEL, all findings are subject to the project-wide $n = 4$ donor constraint (2 CD / 2 Control), compounded here by the donor-specific effect identified above.

Project-Wide Limitations and Considerations

Donor sample size

Both IEL and LPL compartments are limited to $n = 4$ total donors (2 CD / 2 Control). No donor-blocking or paired design has been used in any analytical layer of this project, including this trajectory analysis. Condiments’ donor-aware statistical tests (`topologyTest_multipleSamples`, `progressionTest_multipleSamples`, `fateSelectionTest_multipleSamples`) were considered but not adopted as the primary analytical framework, given that the compositional imbalance documented above makes most branches ineligible for meaningful cross-condition testing regardless of the test’s donor-awareness.

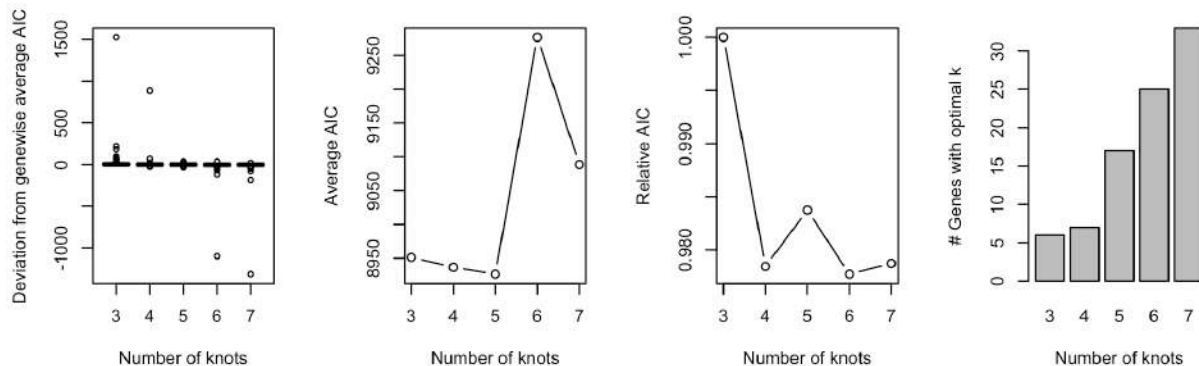
Because trajectory inference and descriptive pseudotime summaries were performed at the cell level, donor-to-donor variability cannot be estimated reliably with only two donors per condition. Consequently, the inferred topology and pseudotime distributions should be interpreted as dataset-level descriptive structures rather than donor-generalized disease effects.

No formal trajectory inference method comparison

Slingshot was used as the sole trajectory inference method. An independent topology cross-validation using a second trajectory inference framework was not performed and therefore represents a methodological limitation of the analysis.

tradeSeq not applied

tradeSeq (`fitGAM` and downstream tests) was attempted on the IEL compartment but was not computationally tractable on standard hardware even after `nknots` selection via `evaluateK` (k range truncated to 3–7 for feasibility; the AIC-relative elbow fell clearly at $k = 4$, within the tested range) and gene-set filtering strategies; a single `fitGAM` run was projected at approximately 1 day 19 hours and was abandoned. No gene-level, model-based differential expression along pseudotime is reported for either compartment. Descriptive, non-model-based marker-trend plots (binned mean expression) are provided as a lower-rigor substitute for the biological questions that tradeSeq would otherwise have addressed.



Interpretive tier framework

Consistent with the validity-tier framework already applied to CellChat/LIANA and pseudobulk DESeq2 elsewhere in this project (fully comparable / imbalanced / CD-emergent / Control-dominant), no trajectory finding in this report should be read as a differential comparison between CD and Control. All condition-related observations are descriptive characterizations of where each condition’s cells sit on a condition-agnostic map, not statistically tested differences in trajectory kinetics.

Reproducibility

All analyses were performed in R using Seurat v5 (LayerData-based assay access), SingleCellExperiment, Slingshot, and condiments (`imbalance_score` only). A fixed random seed (`set.seed(1234)`) was used throughout. Complete session information for both compartments is stored as `sessionInfo_Trajectory_IEL.txt` and `sessionInfo_Trajectory_LPL.txt` in the project’s logs directory; **exact package version numbers should be inserted here from those logs prior to manuscript submission**, rather than restated from memory in this report.

Table 2: Software versions used in the trajectory analysis (placeholders to be completed from saved session logs).

Component	Version
R	4.6.0
Seurat	v5 (LayerData API)
SingleCellExperiment	1.34.0
slingshot	2.20.0
condiments	1.20.0

Conclusion

This trajectory analysis provides a condition-agnostic, descriptive map of T-cell states in both the IEL and LPL compartments of the Jaeger et al. dataset, deliberately avoiding formal CD-versus-Control statistical testing given the severity of condition-associated compositional imbalance and the project's $n = 4$ donor constraint. Two findings arising from careful quality control materially strengthen the project's overall interpretive rigor rather than merely qualifying it: the IEL shared CD4/CD8 trunk is shown, through direct marker verification, not to support a literal reprogramming narrative; and the LPL donor-effect finding provides a concrete, donor-level explanation to a discrepancy with the source literature that had previously been left as an open question in the bulk RNA-seq report. Both findings should be carried forward explicitly into the manuscript rather than treated as analysis-stage-only caveats.